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Variation of Ambient Illumination Across Different Locations: An Impact on Antimyopia Strategy

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Abstract:

PURPOSE: To evaluate whether the level of ambient illumination in winter at different locations, time of a day, sun protection, and source position remains optimum (≥ 1000 lux) for controlling myopia progression.

METHODS: Illuminance level was recorded for 6 outdoor and 5 indoor locations using factory calibrated digital lux meter and under different measurement conditions. Outdoor locations included "open playground," "between two buildings," "between three buildings," "under a porch facing east," "under a big tree," and "under a porch facing west." Similarly, indoor locations included "rooms with multiple large windows," "rooms with a combination of light sources," "room with single artificial light," "room with multiple artificial lights," and "canopy covering the buildings."

RESULTS: The overall median illumination level across 6 outdoor locations and 5 indoor locations were 1900 lux (803–4300 lux) and 227 lux (82–556 lux) respectively. Indoor locations showed an overall low median illuminance level (<1000 lux) irrespective of the locations and source positions. However, the illuminance level >1000 lux was recorded for "Room with multiple large windows" for specific points of time (10:00–11:00 and 13:00–14:00) in both sunny and foggy weather. The highest illumination level was recorded only in the "open playground" irrespective of various conditions such as time of the day, weather of the day, relative source position, and sun protection.

CONCLUSION: The ambient illumination level was optimum (≥ 1000 lux) for antimyopia strategy in winter for maximum outdoor locations. However, children should be motivated to spend their outdoor time during the afternoon (13:00–14:00 h) so that light intensities reaching the eye should be optimum in winter.

Keywords:

Ambient light, antimyopia, illumination, myopia progression, northern India

Introduction

Myopia is the most prevalent refractive error affecting school-going adolescents in both urban and rural regions of India. Being the second most populated country in the world, India holds 42% of the population (492 millions) under 18 years of age.^[1] The increasing prevalence of myopia over the past few decades is

likely due to environmental factors.^[2] A recent systematic review and meta-analysis estimate that the percentage of the world's population with myopia will increase from 22.9% in 2000 to 49.8% in 2050.^[3] Similarly, a systematic review of myopia prevalence in India predicted that the prevalence of myopia in India will increase to 31.89% in 2030, 40.01% in 2040, and 48.14% in 2050.^[4] Various strategies have been employed in an attempt to minimize the myopia progression that includes orthokeratology, bifocal Lens, low dose

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atropine (0.01%), progressive addition lens, aspheric lens, and increased outdoor activities.^[5-10]

Recent studies have indicated time spent outdoors to be protective for myopia or to delay the onset of myopia.^[11-13] While Wu *et al.*^[11] reported that school children with longer outdoor time in school (≥ 200 min) showed significantly less myopic shift, a recent randomized clinical trial showed that the addition of 40 min of outdoor activity at school compared with usual activity resulted in a reduced incidence rate of myopia over the next 3 years.^[12] In addition, recent studies have shown minimum outdoor illumination of 1000 lux to be effective in controlling myopia progression.^[11,13] Similarly, Ho *et al.*^[14] suggested 120 min of daily outdoor exposure during school hours as the most effective intervention in controlling myopia. Studies that quantified the light exposure pattern by estimating illuminance using a light tracker reported that myopic children spent most of their time in an indoor environment with illuminance level <1000 lux every day.^[13,15] In addition, a recent study reported reduction in odds of developing myopia when exposed to an illuminance level of >3000 lux every day.^[16]

There can be several variations of light intensities in outdoor and indoor environment. For example, children playing in open playground can have big trees or open top. Similarly, children playing in indoor environment can have rooms without window or with single or multiple windows. Consequently, the level of illumination in outdoor can also be affected by weather (cloudy/sunny) or seasons (summer/spring/winter). Lanca *et al.*^[17] found greater differences between outdoor (11,080–18,176 lux) and indoor (112–156 lux) locations. In addition, Lanca *et al.*^[17] also reported that the use of sunglasses/hat still provided illuminance (at the eye level) 11–43 times higher than that of indoors. Bhandary *et al.*^[18] found changes in illuminance level under different conditions during winter and summer in India. However, the measurements were done in the southern part of India where the effect of winter does not remain the same as in northern India.

With regard to myopia, previous literatures^[19,20] have reported that the myopia progression was slower in the summer season than winter season due to greater time spent outdoor in summer than winter. In addition, Nioi *et al.*^[21] have reported that the duration of bright light was over 1000 lux for 46 min in summer and 3 min in winter which further suggested greater variation in the light intensities between two seasons.

Considering the fact that the level of optimum illumination and myopia progression varies with seasons (summer/winter), the present study aimed to evaluate whether the level of ambient illumination in winter at different locations, time of a day, sun protection (hat/cap) and

source position remains optimum (≥ 1000 lux) for controlling myopia progression in northern India.

Methods

An experimental study was conducted between 1st week of December 2022 and 1st week of January 2023 in Moradabad, which is the northern region of India. The study was approved by the institutional review board of the institute. Illuminance level in “lux” was measured using factory-calibrated digital lux meter (HTC LX-101 A, Hatco Industries, Mumbai, and Maharashtra, India). The range of the measurement for the lux meter was between 0 and 200000 lux. The measurement was done by a single examiner in 6 outdoor and 5 indoor locations. To ensure the accuracy of the measurements at each location, the display screen was set to zero by switching off the lux meter, and the cap of light sensor was covered as recommended in the manual provided by manufacturer.^[22] The measurement of light intensities was obtained by keeping the light sensor close to the examiner's eye to obtain the closer value of the illuminance level entering the eye. The mean of three readings obtained at each location was noted and was used for further analysis. The methodology of the present study was adopted from previous literature.^[18]

Outdoor and indoor locations

A total of 11 locations were selected for study purpose and were based on the study done in southern India.^[18] There were 6 outdoor locations and 5 indoor locations. Outdoor locations included open playground, between two buildings, between three buildings, under a porch facing east, under a big tree, and under a porch facing west. Similarly, indoor locations included rooms with multiple large windows, rooms with combination of light sources, room with single artificial light, rooms with multiple artificial lights, and canopy covering the buildings. The locations were selected keeping in mind the places that children were likely to spend their time for the day. The description of the locations is represented in Table 1.

Time of the day, source positions, weather, and sun protection

The illuminance level was measured along four different time points in a day (7:00–8:00, 10:00–11:00, 13:00–14:00, and 16:00–17:00) on foggy and sunny days. The forecast of the day (as sunny, cloudy, or foggy) was recorded from the default weather application on an IOS smartphone (AccuWeather application, <https://www.accuweather.com/en/in/moradabad/191056/weather-forecast/191056>). Similarly, the measurements in outdoor locations were obtained by positioning the sensor of the lux meter in three different directions relative to the source of light, i.e. (i) facing toward the source (TS), (ii) opposite from source (OS) and (iii)

Table 1: List of 6 outdoor and 5 indoor locations with their descriptions

Locations	Description
Outdoor locations	
Open playground	An open area of land where children used to play
Between two buildings	A hallway covered with two tall buildings from two sides
Between three buildings	An open space covered with three tall buildings from three sides
Under a porch facing east	Small covered area from one side with a tall building facing toward west
Under a big tree	Big tree in playground
Under a porch facing west	Small covered area from one side with a tall building facing toward west
Indoor locations	
Room with multiple large windows	A large room with two large glass windows facing toward east and north. (Dimensions in feet - 36×25×11.5)
Room with combination of light sources	A room with two large glass windows facing toward east and north with ceiling LED lights. (Dimensions in feet - 36×25×11.5)
Room with single artificial light	A closed room with a single ceiling LED source. No natural light source. (Dimensions in feet - 17×10×9.5)
Room with multiple artificial lights	Large room with multiple ceiling LED sources. No natural light source. (Dimensions in feet - 41×42×10)
Under a canopy covered by four buildings	An area covered with four tall buildings on four sides with translucent artificial shade

intermediate to the source which is 90° midway from TS and OS. In addition, the measurement of illumination level for indoor locations was done in two different scenarios, i.e., (i) under the source (the sensor of the lux meter facing toward the ceiling where artificial light was placed) and (ii) away from the source (measurement usually 1–2 m away from the source). For outdoor measurements, the illuminance levels were also measured using sun protection (hat and cap) to evaluate the influence of illuminance level reaching the eye.

Statistical analysis

Statistical analysis was performed using SPSS software (IBM SPSS, version 25; IBM Corp., Armonk, NY, USA). The term “overall” was defined for the illuminance level recorded for all conditions (time of the day, with and without sun protection, position of sensor relative to source, and weathers) for each location. Data were represented in mean and interquartile range (IQR) for different outdoor and indoor locations.

Results

Overall illuminance level for outdoor and indoor locations

The overall median illumination level across 6 outdoor locations and 5 indoor locations were 1900 lux (803–4300 lux)

and 227 lux (82–556 lux). The overall median illuminance level for outdoor locations was 8.3 times more than indoor locations. Considering each location for outdoors, the highest overall illumination was recorded in “open playground” (median: 5800; 2700–11800 lux) followed by “between two buildings” (median: 2200; 1100–3500 lux) and “under a porch facing west” (median: 2150; 941–3200 lux). However, the median illuminance level for “between three buildings” and “under a porch facing east” was less than half of the illuminance level for “open playground.” Similarly, the lowest overall illumination level was recorded in “between 3 buildings” (median: 1179; 442–2900 lux) followed by “Under a big tree” (median: 1200; 600–2300) and “under a porch facing east” (median: 1400; 688–2800 lux). Though all the outdoor locations had recorded optimum illumination (≥ 1000 lux), the three locations (under a porch facing east, between three buildings and under a big tree) had recorded median value very close to the optimum level of illumination [Table 2].

Indoor locations showed an overall low median illuminance level (< 1000 lux) irrespective of the locations and source positions. However, the illuminance level more than 1000 lux was recorded for “Room with multiple large windows” for specific point of time (10:00–11:00 and 13:00–14:00) on both sunny and foggy weather. The highest level of illuminance was recorded for “canopy covered with four tall buildings” (median: 628; 345–1125) followed by “room with multiple large windows” (median: 573; 290–1200) [Table 2].

Illuminance level in sunny and foggy days

The overall median illuminance levels recorded in sunny days and foggy days were 2600 lux (1200–5500 lux) and 1400 (620–3300 lux). The overall illumination noted for foggy days of winter season was almost half of the illuminance level of sunny days. Only one outdoor location (open playground) showed an illuminance level > 1000 lux irrespective of the conditions (time of the day, sun protections, and relative position of sensor). In addition, the outdoor locations that recorded illuminance levels ≥ 1000 lux under direct source of light for all the timings of the day were “open playground” (median: 10500; IQR 10000–11200 lux), “between two buildings” (median: 7500; IQR 3500–15000 lux), “between three buildings” (median: 4200; IQR 2300–7500 lux), “under a porch facing east” (median: 3650; IQR 1800–5850 lux), and “under a porch facing west” (median: 4350; IQR 2700–7000 lux) in sunny days [Figure 1] while only three outdoor locations recorded illuminance level ≥ 1000 lux in foggy days that included “open playground” (median: 9250; IQR 5300–10600 lux), “Between two buildings” (median: 4650; IQR 2125–7575 lux), and “Between three buildings” (median: 3150; IQR 1700–6275 lux) irrespective of all conditions [Figure 2].

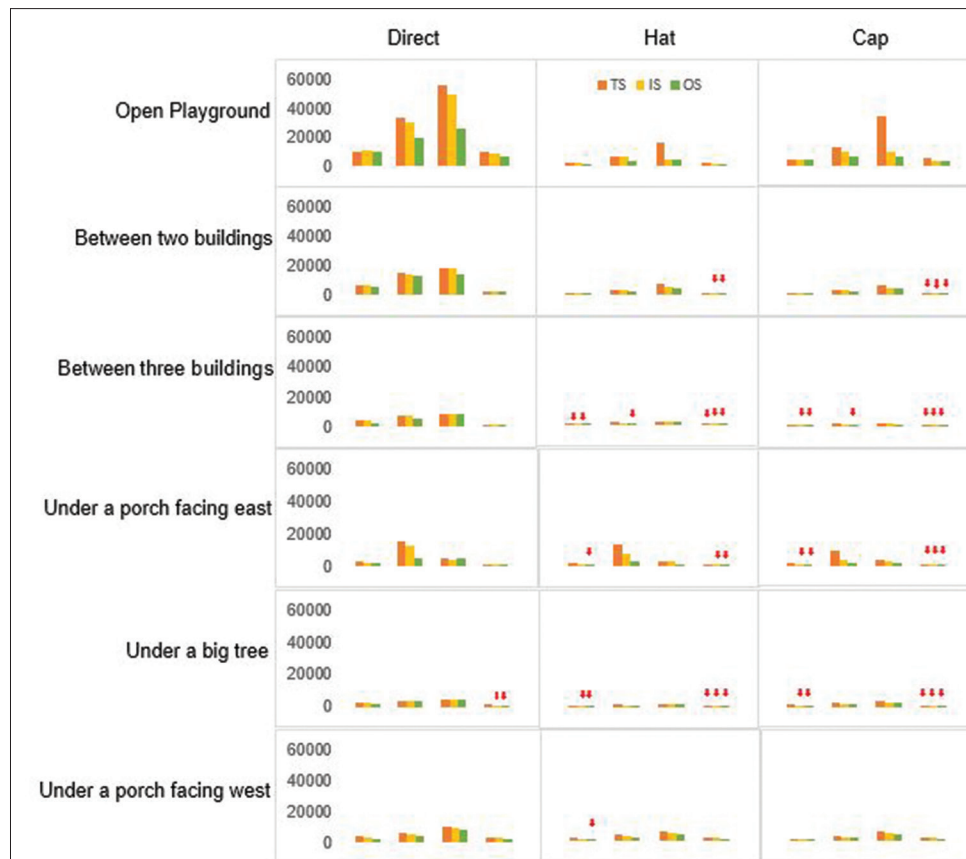


Figure 1: Median illuminance level recorded under direct source, hat, and with a cap on 9 sunny days. Toward the source, intermediate to the source, and opposite from source sensor source positions recorded in four different time points of a day (07:00–08:00, 10:00–11:00, 13:00–14:00, and 16:00–17:00 h). The downward-facing arrows in red color represent illuminance levels <1000 lux

Table 2: Overall illuminance level for 6 outdoor and 5 indoor locations

Locations	Overall illuminance level (lux)	
	Median (IQR)	Range
Outdoor locations		
Open playground	5800 (2700–11,800)	270–88,400
Between two buildings	2200 (1100–3500)	99–50,800
Between three buildings	1179 (442–2900)	26–17,000
Under a porch facing east	1400 (688–2800)	71–46,200
Under a big tree	1200 (600–2300)	103–12,500
Under a porch facing west	2150 (941–3200)	110–13,000
Indoor locations		
Room with multiple large windows	573 (290–1200)	160–30,800
Room with combination of lights sources	418 (86–1345)	12–31,000
Room with single artificial lights	79 (24–220)	10–490
Room with multiple artificial lights	320 (195–430)	50–710
Canopy with four buildings	628 (345–1615)	117–3300

IQR: Interquartile range

weather in indoor location's illuminance level. All the indoor locations recorded the illuminance level <1000 lux irrespective of all measurement conditions except two indoor locations (room with multiple large windows and room with combinations of light sources) that had recorded >1000 lux between 13:00 and 14:00 h [Figure 3].

Illumination level at different points of time for the day

The median illuminance level of all outdoor locations was ≥ 1000 lux during 13:00–14:00 h (median: 1900; IQR 1400–5800 lux) followed by 10:00–11:00 h (median: 1650; IQR 1100–5625 lux). Similarly, “room with multiple large windows” (median: 5500 IQR 1670–11350 lux) and “Room with combination of light sources” (median: 5200; IQR 3075–9000 lux) were the two indoor locations that had recorded the illuminance level ≥ 1000 lux between 13:00 and 14:00 h for both sunny and foggy weathers for measurement under direct source [Figure 3].

Discussion

In addition, all the outdoor locations showed illuminance level ≥ 1000 lux between 13:00 and 14:00 h of the day irrespective of all the measurement conditions in sunny and foggy weathers. However, there was no effect of

The present study evaluated the median ambient illumination level in different outdoor and indoor locations in winters in northern India for deciding

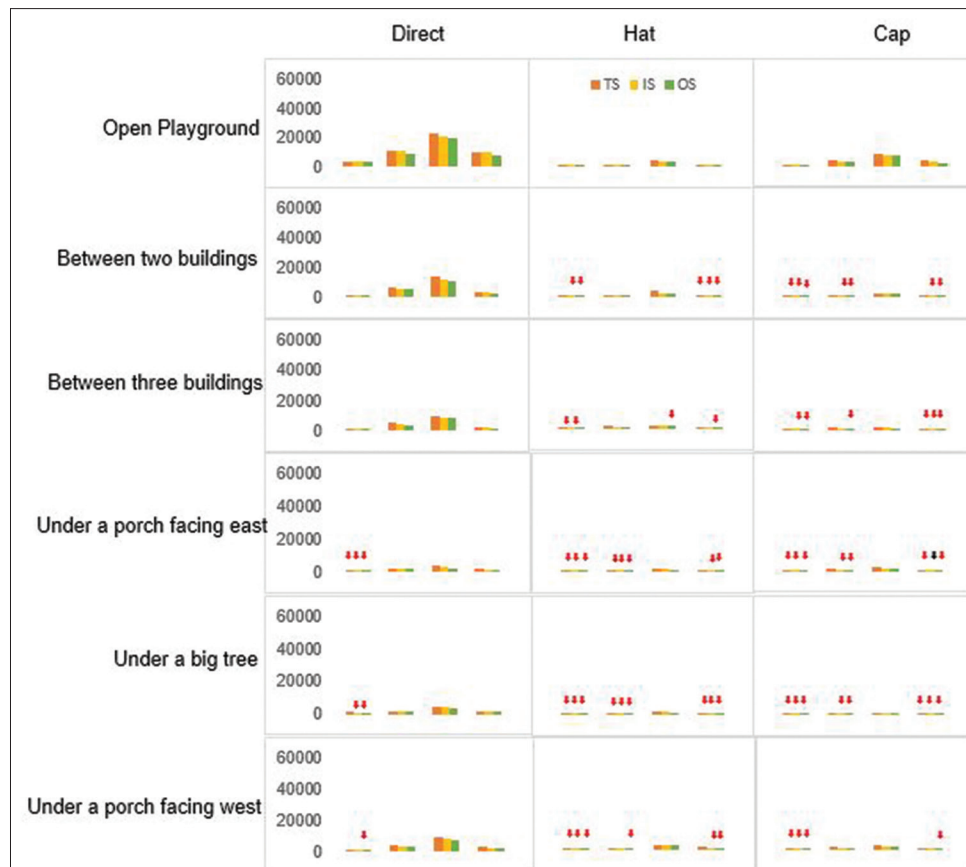


Figure 2: Median illuminance level recorded under direct source, hat, and with a cap on 11 foggy days. Toward the source, intermediate to the source, and opposite from source sensor source positions recorded in four different time points of a day (07:00–08:00, 10:00–11:00, 13:00–14:00, and 16:00–17:00 h). The downward-facing arrows in red color represent illuminance levels <1000 lux

optimum conditions to prevent myopia progression. The current study found that the highest illumination level remains only in “open playground” irrespective of various conditions such as time of the day, weather of the day, relative source position, and sun protections. Similarly, the average median illuminance level of the indoor locations was <500 lux and was about 8.3 times less than outdoor locations. In addition, we found that the illuminance level was ≥ 1000 lux for all outdoor locations between 13:00 and 14:00 h irrespective of the weathers and source positions.

Low level of illuminance were recorded in all indoor locations and measurement conditions with median illuminance <500 lux showed agreement with previous studies.^[11,17,23] Lanca *et al.*^[17] reported that outdoor light level was approximately 100 times more than indoor light levels (11,080–18,176 lux vs. 112–156 lux during cloudy days in Singapore. Similarly, Bhandary *et al.*^[18] also found that the illuminance level of outdoor locations was 8 times more than indoor locations irrespective of the different measurement conditions in Hyderabad, India.

The overall median and range of illuminance values of all outdoor locations observed in the current study is lower

than that reported in Hyderabad^[18] which could be due to differences in the seasons and location of the study (we measured illuminance level on sunny and foggy days of winter season in northern region of India where the average temperature was 16°C and 8°C, respectively, while Bhandary *et al.*^[18] measured illuminance level on sunny and cloudy days in southern region of India where the average temperature remained greater than northern region).

While five locations in this study showed a median high illuminance of ≥ 1000 lux in sunny days of winter, only three outdoor locations, namely “between two buildings,” “between three buildings,” and “Open playground” recorded optimum illuminance levels, irrespective of all conditions. Previous studies have indicated that illuminance level >1000 lux itself might have beneficial effects in either preventing myopia or delaying the onset of myopia.^[24] In addition, Wu *et al.*^[11] reported that exposure to high illuminance (strong sunlight) may not be necessary for myopia prevention and instead indicated that spending greater time (close to 200 min per day) in locations with illuminance levels > 1000 lux may be sufficient to slow the progression or prevent myopia. Therefore, the potential of any indoor, artificial, or seminatural locations that

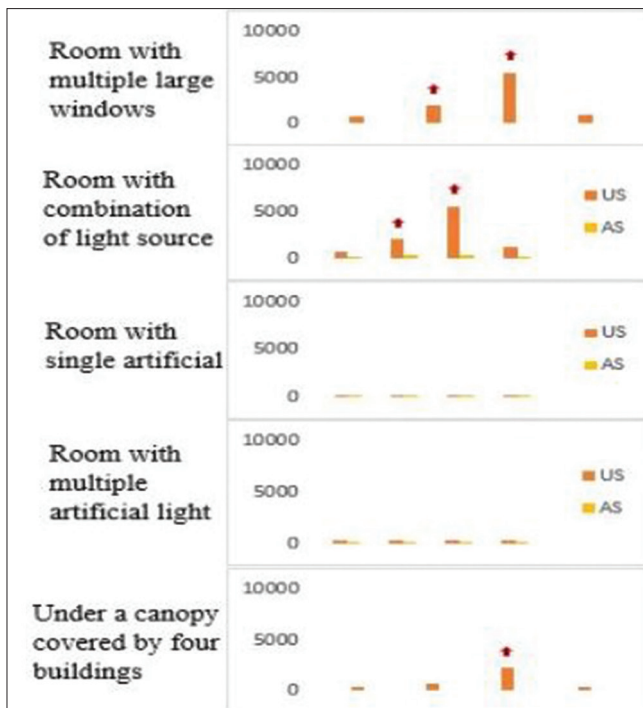


Figure 3: Median illuminance level recorded in five indoor locations on 11 foggy and 9 sunny days. US and AS sensor source positions recorded in four different time points of a day (07:00–08:00, 10:00–11:00, 13:00–14:00, and 16:00–17:00 h). The upward-facing arrows in red color represent illuminance levels >1000 lux

has illuminance levels >1000 lux cannot be ruled out for myopia control in humans.^[25]

Hébert *et al.*^[26] evaluated the diurnal distribution of ambient illumination. Time spent awake was divided into four-time intervals: morning (wake time to 12:00), afternoon (12:00–16:00), early evening (16:00–20:00), and late evening (20:00 to bedtime). Except for late evening, the time spent in bright illumination was significantly longer during the summer for all time intervals, but the relative distribution of bright light exposure throughout the day was the same in both seasons. In addition, subjects spent more than 50% of their time dimmer than 100 lux irrespective of seasons. The current study recorded illuminance level >1000 lux between 13:00 and 14:00 h irrespective of the different measurement conditions. Wu *et al.*^[11] found that students with longer outdoor timings with illumination level >1000 lux showed less myopic shift. Similarly, Read *et al.*^[13] also noted that children with low average daily light exposure exhibited significantly greater annual eye growth than moderate and high daily light exposure. In support to previous literatures,^[11,13] children should be motivated for outdoor activities in the afternoon (12:00–16:00 h), specifically in winter season which could restrict the myopia progression in children.

Donovan *et al.*^[19] found that the mean 6-month spherical equivalent (SE) progression and mean axial elongation

varied with different seasons (summer, autumn, spring, and winter). However, winter and summer seasons showed significant differences in axial length elongation and SE progression. Similarly, Gwiazda *et al.*^[20] also found the variation of myopia progression with months; slower in April–September than other months. Mean progression in winter was -0.35 ± 0.34 D and in summer was -0.14 ± 0.32 D.

Taking sunny and foggy weather into consideration, we found the illuminance level recorded for sunny days was almost twice the illuminance level for foggy days. Considering the findings of Bhandary *et al.* and present study, we found the level of illuminance was greatly affected by the seasons and geographical locations (e.g. “open playground” recorded overall median illuminance level recorded was 9300 lux in Bhandary *et al.*’s study while 5800 lux in the present study). Simultaneously, the probable reason for slower progression of myopia in summer season could be due to the high illuminance level reaching to the eye, compared to winter seasons with constant outdoor timing. However, the study was conducted in two different regions of India that have diversified weathers and seasons. Further investigations are warranted in the northern part of India to compare the illuminance level between winter and summer and role of illuminance in myopia progression.

The greatest strength of this study is that we investigated illuminance levels in 6 outdoor and 5 indoor locations which are common/regular places where children are likely to be spending their time. Similarly, the study was limited by the following conditions:^[1] only data of the winter season were collected.^[2] Illuminance level was recorded only in Moradabad city, which is in the northern part of India, but the geographical spread of India is wide where the weather and season can vary with different states.

Conclusion

Open playground was the only outdoor locations with illuminance level of >1000 lux that could minimize the myopia progression in the winter season in North India irrespective of all measurements. Children should be motivated to spend their outdoor time during afternoon (13:00–14:00 h) so that light intensities reaching to the eye should be optimum. Finally, keeping in view the variation in illuminance in different locations, children and parents need to be recommended optimum time outdoors as antimyopia strategy.

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Conflicts of interest

There are no conflicts of interest.

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